

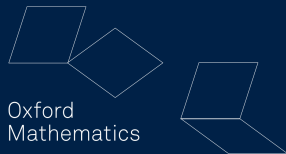
# Differentiable Electrochemical Simulation for Gradient-Based Parameter Inference



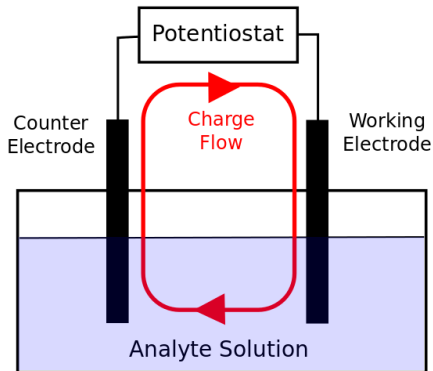
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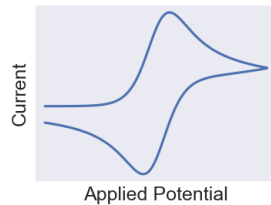
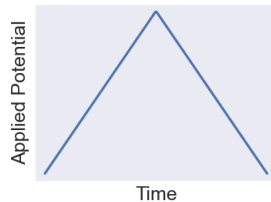
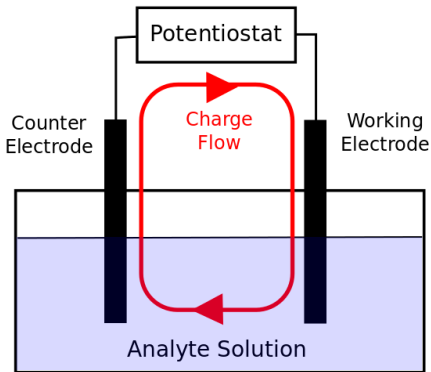
OMMS Dissertation Presentation



# Electrochemistry and Cyclic Voltammetry



# Electrochemistry and Cyclic Voltammetry



# Mathematical Description

► **Diffusion:**

►  $\frac{\partial c_A}{\partial t} = D_A \frac{\partial^2 c_A}{\partial x^2}$

►  $\frac{\partial c_B}{\partial t} = D_A \frac{\partial^2 c_B}{\partial x^2}$

► **Initial Condition:**  $c_A(0, x) = c_A^*$ ,  $c_B(0, x) = c_B^*$

► **Infinite Boundary Conditions:**

►  $c_A(t, x) \rightarrow c_A^*$  as  $x \rightarrow \infty$

►  $c_B(t, x) \rightarrow c_B^*$  as  $x \rightarrow \infty$

► **Electrode Boundary Condition (Butler-Volmer):**

►  $D_A \frac{\partial c_A}{\partial x} \Big|_{x=0} = k_0 \left[ c_A(t, 0) e^{-\alpha \frac{F}{RT} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{RT} (E(t) - E_f^0)} \right]$

►  $D_B \frac{\partial c_B}{\partial x} \Big|_{x=0} = -D_A \frac{\partial c_A}{\partial x} \Big|_{x=0}$

► **Current:**  $I = FAj_{0,A}$  where  $j_{0,A} = -D \frac{\partial c_A}{\partial x} \Big|_{x=0}$

# Mathematical Description

Assuming  $D_A = D_B$

► **Diffusion:**  $\frac{\partial C_A}{\partial t} = \frac{\partial^2 C_A}{\partial x^2}$

► **Initial Condition:**  $C_A(0, X) = 1$

► **Infinite Boundary Conditions:**  $C_A(t, X_{\max}) = 1$

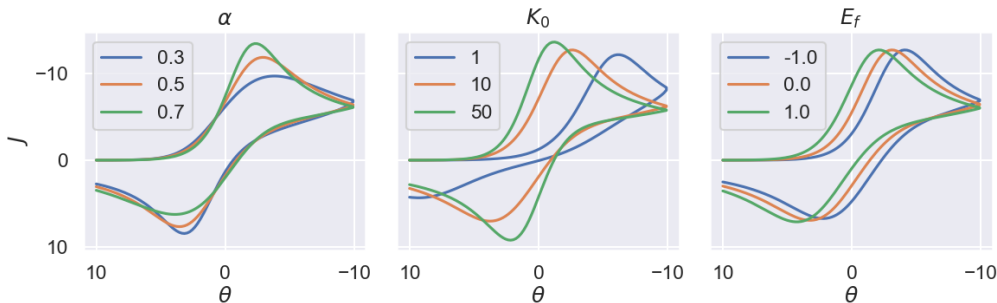
► **Electrode Boundary Condition (Butler-Volmer):**

$$\left. \frac{\partial C_A}{\partial X} \right|_{X=0} = K_0 \left[ C_A(T, 0) e^{-\alpha(\theta(T) - \theta_f^0)} - (1 - C_A(T, 0)) e^{(1-\alpha)(\theta(T) - \theta_f^0)} \right]$$

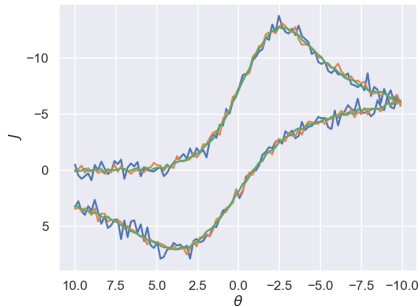
► **Current:**  $I = \pi \epsilon F D_A c_A^* J_{0,A}$  where  $J_{0,A} = -D \left. \frac{\partial C_A}{\partial X} \right|_{X=0}$

# Forward Problem

- Set  $\phi = (\alpha, K_0, \theta_f)$
- Finite Difference Solver:  $\mathcal{P}(\phi) = J$



# Inverse Problem



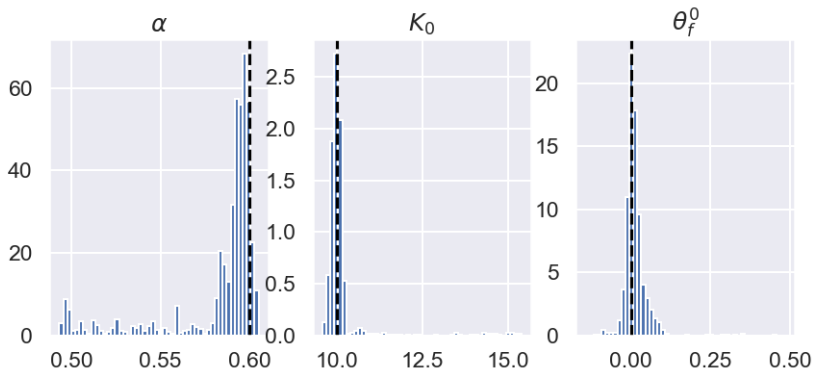
$\phi$

- ▶ Aim to approximate the posterior distribution  $P(\phi|y)$ , where  $y$  is the experimental data
- ▶ Based on simulating a Markov chain whose limiting distribution is the posterior distribution
- ▶ Parameters are proposed based on a proposal distribution  $q(\phi_{\text{cand}}|\phi)$
- ▶ Proposals are accepted with probability:

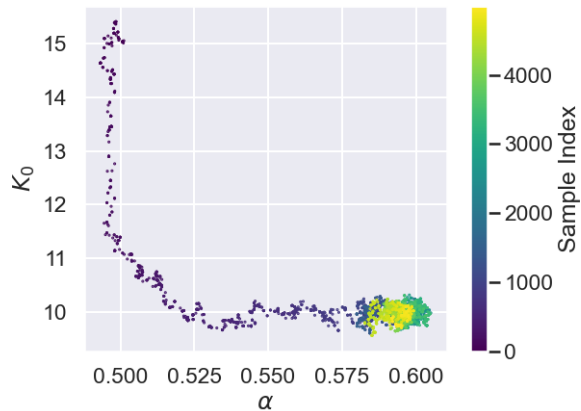
$$r = \min \left\{ \frac{L(\phi_{\text{cand}}|y)}{L(\theta_i|y)}, 1 \right\}$$

# Random-Walk Metropolis Hastings

$$q(\theta|\theta_t) \sim \mathcal{N}(0, \Sigma)$$



# Burn-in

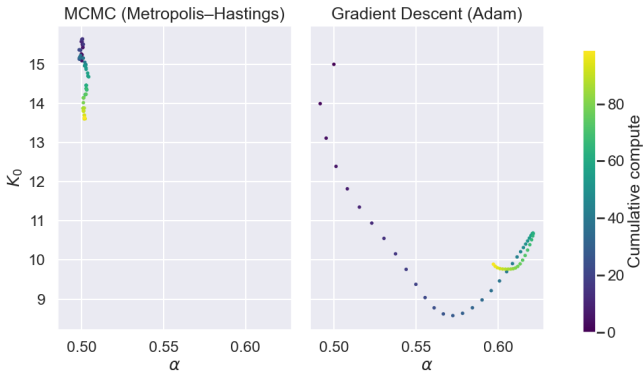


# Differentiable Simulator

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- ▶  $\mathcal{P}$  is a composition of differentiable functions so it is itself differentiable.
- ▶ Hard to derive analytically, but we can compute the derivatives efficiently using automatic differentiation
- ▶ Compute  $\Delta_\phi L$

Performing gradient descent on the likelihood can help us enter areas of high density



faster.

# Hamiltonian Monte Carlo

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# Further Work

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